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Introduction

Studies suggest that 39% percent of first-time students started but did not graduate higher education (Education Data Initiative, 2025). According the Federal Reserve, 28 percent of people who attended some but never completed college education had student debt (federalreserve.gov, 2025). Depending on the particular field, a college degree can serve as a proxy for revelant job skills and used to screen the applicant pool. A 2022 study from the Burning Glass Institute and Harvard Business School found that 37% of middle skilled jobs still require a college. This puts these individuals at a potential economic disadvantage, as some may find that they don't meet the expected qualifications in certain job markets while also having to worry about paying back their debt. In fact, college dropouts earn an average of 35% less income than graduates. This can make it difficult to find their way back to school or to gain upward mobility. In addition to the costs to individuals, the effects of dropouts can have a negative impact on businesses as well. While some employers are experimenting with resetting/lowering their degree requirements, those that still have degree requirements risk turning away highly skilled candidates that could restrict the growth potential of the business and its broader sector (Burning Glass Institute, 2022).

This study aims to train a Machine Learning (ML) model to help identify student's at-risk of dropping out of college. From a dataset from Kaggle, I will explore the importance of 34 variables covering topics such as student's personal profile (i.e. marital status, gender, age, debt), their school profile (i.e. enrollment type, major), academic achievement over two semesters (i.e. number of courses attempted and passed), parent's educational level and occupation, and local economic conditions (i.e. unemployment, inflation rate, and GPD). I will train and evaluate models using the Decision Trees, XGBoost and Neural Networks ML algorithms to identify the strongest predictive model. As a secondary objective, the selected models will be used to answer *what are the factors that contribute to dropouts?* with the hope that it can inform future interventions that improve the economic outcomes for students and businesses alike.

1. EDA and Data Engineering

The dataset is comprised of 4424 samples with 16 numerical variables, 18 categorical variables including our dependent variable `target`. Each sample represents a unique student record for one year (2 semesters) of college participation with an outcome (Dropout, Graduate and Enrolled) assigned to `target`. Roughly 18% of our observations were categorized as **Enrolled**, meaning that the student is still in school does not have a final outcome (graduated vs dropped out). For the purposes of this study, I have held out these observations to use as my inference set to predict the likely final status, as training my models with this class might make obscure the signals for predicting dropouts and graduates. 80% of the remaining 3467 observations will be used to train my models, while 20% will be used to test them.

|———-|——|——-| | Graduate | 2129 | 50.31% | | Dropout | 1338 | 38.59% |

The dataset has no missing (N/A) values. However, 180 observations (~4%) had a value of 0 in all twelve fields for academic achievements in the last two semesters. These fields include valuable information about the number of courses attempted, passed, and credits awarded per semester, as well the corresponding number of tests and grades. Figure 1 shows the boxplots for our continuous variables demonstrating that students that dropout are more likely to have received lower grades and passed fewer courses than those that graduated or those still enrolled. The records in question had mixed outcomes in the `target` field, where some were enrolled and others were graduates or dropouts making it difficult to identify a particular pattern for why this value was used. This suggests that 0 could be a stand-in for a missing value possibly from previous imputation, that the dataset was synthetically generated, or that these records could be representative of edge cases (ie students that completed/were enrolled in course work in a previous year but whose outcome status was not updated until the year of observation), as it seems highly unlikely that a student could be a graduate without taking at least one course. Given the connection between low grades and failed courses with dropouts and that we have no way of verifying these records, I opted to drop them over performing imputation as to avoid introducing noise into our model.

Our boxplots also show that the IQRs for `unemployment_rate` and `application_order` are practically identical for the Dropout and Graduate classes, suggesting that these variables may not add much information to our models. I omited these two variables from my model so as to help reduce the overall dimensions used by my models and improve training efficiency.

Examining our categorical variables show other areas where we may be able to bin categories to reduce dimensions created by One-Hot encoding. In particular, `application_mode`, `previous_qualifications`, mother and father's "qualification" and "occopation" fields, and `nationality` all have multiple categories of few observations (less than 5%) that can be grouped together to keep the number of dimensions from growing too high.

Testing for Pearson Correlation Coefficient and Variance Inflation Factor (VIF) shows high correlation between 1st and 2nd semester academic achievement fields as seen in Table 1 and Table 2. I created new engineered variables combining some of these fields to help minimize correlation and as a dimension reduction method. The fields created were:

- `total_enrolled`: the sum of `curricular_units_1st_sem_enrolled` and `curricular_units_2nd_sem_enrolled` represeing the total number of courses that the student was enrolled in during the school year;

- `total_credited`: the sum of `curricular_units_1st_sem_credited` and `curricular_units_2nd_sem_credited` representing the total number of credits received during the school year;
- `total_approved`: the sum of `curricular_units_1st_sem_approved` and `curricular_units_2nd_sem_approved` representing the total classes passed during the school year;
- `total_evaluations`: the sum of `curricular_units_1st_sem_evaluations` and `curricular_units_2nd_sem_evaluations` representing the total number of courses with tests during the school year;

Finally, I also created an engineered field for the students weighted average grade (`weighted_avg_grade`) and ommited the `curricular_units_[]num_sem_without_evaluations` fields as these fields are complimentary to the `curricular_units_[num]_sem_evaluations` fields and thus contain reduntant information.

3. Model Selection

I used three different Machine Learning techniques to evaluate their predictive performance for Accuracy, Precision, Recall, F1 Score, AUC, AUPRC across 6 models, as shown on Table 1. While all six metrics are important, Recall is more important than Accuracy for my analysis, given that our goal is to capture the highest number of at-risk students F1 Score is also important as a high F1 Score it is the harmonic will balances precision and recall, ensuring that we prioritize both.

I trained a Decision Tree with a max depth of 5 as my baseline model due to the high interpretability of Decision Trees. Although this performed decently in Accuracy and Precision, Decision Trees are also highly prone to high variance and overfitting since small changes in the training data can produce different splits.

I proceeded by experiment with XGBoost, an ensemble learning method that builds many shallow trees sequentially to adjust errors resulting in a more accurate and efficient model with slightly higher bias but much lower variance than a single tree, thus preventing overfitting. After training an initial model using random hyperparameters (`XG0`), I manually tested different values for `max_depth`, `min_child_weight`, `gamma`, `n_estimators`, and `learning_rate` with 5-fold CV to create model `xg1`. I them proceeded to use Random Search to perform some hyperparemeter tuning using 10-fold Cross-Validation (CV) and applied the best results to `XG2`. All three models performed similarly with `XG0` outperfoming the others on Recall.

For my final set of experiments, I trained Neural Networks (NN) models. A neural network is a machine learning model that learns patterns from data by passing information through layers of connected nodes. Bias can high for Neural Networks if the network is too small or poorly tuned; Variance can slow for large networks without regularization. I tested Neural Networks with 1 and 2 hidden layers and compared different input and output dimensions as well as different learning rates. Figures 3.4-3.7 shows charts comparing Loss and F1 Scores for Neural Networks with 1 and 2 hidden layers. `NN1` represents the best perfoming model with a single layer of dimension [32,1] with a learning rate of 1e-3. `NN2` represents the best perfoming model with two layers —— the first with of dimension [64,32] and the second [32,1] – with a learning rate of 1e-4. Both models were trained using the Adaptive Moment Estimation (Adam) optimizer; a weight decay (L2 regularization) of 1e-4 was applied to both models to help prevent the model from overfitting.

Table 1: Comparison of Performance Metrics Across All Models

Model	Accuracy	Prec	Recall	F1	AUC	AUPRC
DTree	0.880747	**0.951456**	0.728625	0.825263	0.892428	0.87016
XG0	0.905172	0.931915	**0.814126**	0.869048	0.932589	0.936413
XG1	0.903736	0.93913	0.802974	0.865731	0.938457	0.939775
XG2	0.905172	0.939394	0.806691	0.868	0.938379	0.938942
NN1	0.905172	0.935622	0.810409	0.868526	**0.940599**	0.940281
NN2	**0.906609**	0.935897	**0.814126**	**0.870775**	0.939911	**0.940774**

Based on our Performance, model `NN2` had the highest score for four of our performance metrics for Accuracy, Recall, F1 Score, and AUPRC, and performed relatively well for Precision and AUC. However, `NN1` and the three XGBoost models performed only slightly below our best numerical performer. XGBoost models are more robust to high bias and high variance than Neural Networks. As an ensemble model comprised of multiple trees, XGBoost models are also considered more interpretable than Neural Network. While all XGBoost models performed well, hyperparameter tuning on a small dataset like the one in our analysis can overfit to the training data and hurt generalization. Therefore, `XG0` may be preferable to its tuned counterparts. Finally, as the goal of our analysis is to inform who to target interventions to prevent dropouts, having higher recall is of high importantance when evaluating our model; `XG0` tested equally to `NN2` in recall and has compiratable performance across the other metrics indicating that it is a well-balanced model.

Conclusion

Given the likely audience of our analysis and our secondary objective of identifying the factors that put students at risk of dropping out, the small performance advantage of `NN2` may not be enough to select this model for production overour XGBoost models. XGBoost models tend to be fairly robust to over- and under-fitting while being more interpretable than Neural Neworks. Of the three XGBoost models testest, I would select `XG0` given that it performed equally as well as `NN2` for Recall and nearly as well across the other metrics while posessing these two inherent advantages over NN2.

Turning our attention to our secondary objective, I used SHAP to answer the question *what are the factors that contribute to dropouts?* . SHAP computes the contribution of each feature to each prediction. Figure 4.1 shows the absolute mean SHAP value ranking feature by model contribution while Figure 4.2 shows the direction and magnitude that it pushes our prediction towards a positive (dropout = 1) or negative class (graduate = 0). The chart shows that being a scholarship holder have the biggest contributor on students graduation, followed by International status and if a student's father is in an "Intermediate Level Profession". Conversly, being "male" and if father's occupation is the armed forces or installation, machine operator, and assembly worker or other appear to contribute more to dropouts, along with the selection of student major. Interestingly, none of our academic achievement for the year examined parameters were included on the chart, suggesting lower overall importance to the model.

Feature importance can be used to develop interventions to help at-risk students reorient student outcomes. For example, the plot shows that scholarship_holders are more likely to affect graduation levels, suggesting that providing additional scholarship and tuition aid opportunities could help at-risk students or extended financing opportunities when students fall behind their tuition payments. Interventions could also include recruiting students at-risk into shorter skill-based programs designed to meet business needs helping to meet potential employment demands by increasin g the pool of trained candidates, serving as substitute for a degree credential, and expanding career opportunities for students. This may be particularly effective if businesses in these sectors can provide income opportunities to help pay for student tuition fees, such as scholarships and paid internships.

There are various limitations of our results. Our model was trained on a small dataset with a limited number of variables that do not represent the full gamut of factors that contribute to a students experiences and decision to leave higher education. For example, the dataset does not consider health factors, wether the student is currently employed, income/assets, monthly expenses or other familial responsibilities that may make it difficult for a student to succeed at school. The results from our findings are most relevant to the population from which they were sampled as factors change over time. It is important emphasize the effects of model decay, whereby the predictive performance will degrade over time as the importance of factors shift. Therefore, our model should be evaluated regularly to ensure with new data samples to ensure that it is capturing timely data. This will may also help make our model more robust as our model may benefit from additional datapoints that can help it generalize to new patterns.

References

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